GlobalMart Dimensional Model – Gold Layer Design Document

1. Overview

The Silver layer gives GlobalMart **clean, standardized**, and **quality-checked** data.
The Gold layer must transform that data into a **dimensional model** designed specifically to address:

GlobalMart's 3 core business failures:

1. **Revenue audit inconsistencies**
2. **Returns fraud visibility gaps**
3. **Inventory blindspots across regions**

And support key business questions:

- Monthly revenue by region
- Vendor return rates
- Customer return history across regions
- Product performance by region
- Most/least profitable segments
- Return patterns per customer

To achieve this, the Gold layer uses:

- **Two Fact Tables**
- **Five Dimension Tables**
- **Clear grains**
- **Consistent business keys**
- **Star schema optimized for BI + AI search**

2. Fact Tables (with Grain, Measures, Business Process)

GlobalMart requires **two fact tables**, each representing exactly one business event:

FACT TABLE 1 — fact_transactions

Business Process:

Product sold as part of an order.

Grain (VERY IMPORTANT):

1 row = 1 line item on an order
(This is the correct grain because revenue, quantity, discount, and profit all occur at the line level.)

Measures:

- `sales_amount`
- `quantity`
- `discount_percent`
- `profit_amount`
- `net_sales` (derived measure: sales − discount)

Foreign Keys:

- `customer_id` → dim_customers
- `product_id` → dim_products
- `vendor_id` → dim_vendors
- `order_date_key` → dim_dates
- `region` → dim_regions

Business Failures Addressed:

Revenue audit

- Detects duplicated order lines
- Supports customer identity standardization
- Tracks vendor and product-level financials

Inventory blindspot

- Source of quantity sold per region
- Shows demand velocity

FACT TABLE 2 — fact_returns

Business Process:

A customer returns a product/order.

Grain:

1 row = 1 return event per order
(If an order is returned, one record exists regardless of number of items.)

Measures:

- `refund_amount`
- `days_to_return`
- `return_flag`
- `high_value_return_flag`
- `fraud_indicator_score` (for AI layer)

Foreign Keys:

- `order_id` → fact_transactions
- `customer_id` → dim_customers
- `product_id` → dim_products
- `return_date_key` → dim_dates
- `region` → dim_regions

Business Failures Addressed:

Returns fraud

- Customer return frequency
- Cross-region return behavior
- High-value return patterns

Revenue audit

- Ensures refunds match valid transactions

3. Dimension Tables (with Attributes & Keys)

These dimensions provide all business context needed by the facts.

DIM_CUSTOMERS

Primary Key: `customer_id`

Attributes:

- customer_name
- customer_email
- segment (Consumer/Corporate/Home Office)
- region
- city
- state
- Postal_code

Built From (Silver Table)

Customers_silver

Supports:

- Customer profitability
- Cross-region fraud detection
- Segmentation analysis

DIM_PRODUCTS

Primary Key: `product_id`

Attributes:

- product_name
- category
- sub_category
- brand
- upc
- size / weight / dimensions

Built From (Silver Table)

products_silver

Supports:

- Product performance
- Slow-moving analysis
- Vendor attribution

DIM_VENDORS

Primary Key: `vendor_id`

Attributes:

- vendor_name
- vendor_type (optional)

Built From (Silver Table)

Vendors_silver

Supports:

- Vendor return rate

- Vendor performance by region

DIM_DATES

Primary Key: `date_key`

(YYYYMMDD)

Attributes:

- full_date
- day
- week
- month
- quarter
- year
- fiscal variants

Built From (Silver Table)

Derived — created as a standard Date dimension, not from raw data

Supports:

- Month-over-month revenue
- Seasonal return trends
- Inventory velocity

DIM_REGIONS

Primary Key: `region`

(East, West, Central, North, South)

Attributes:

- region_name
- country (always USA)

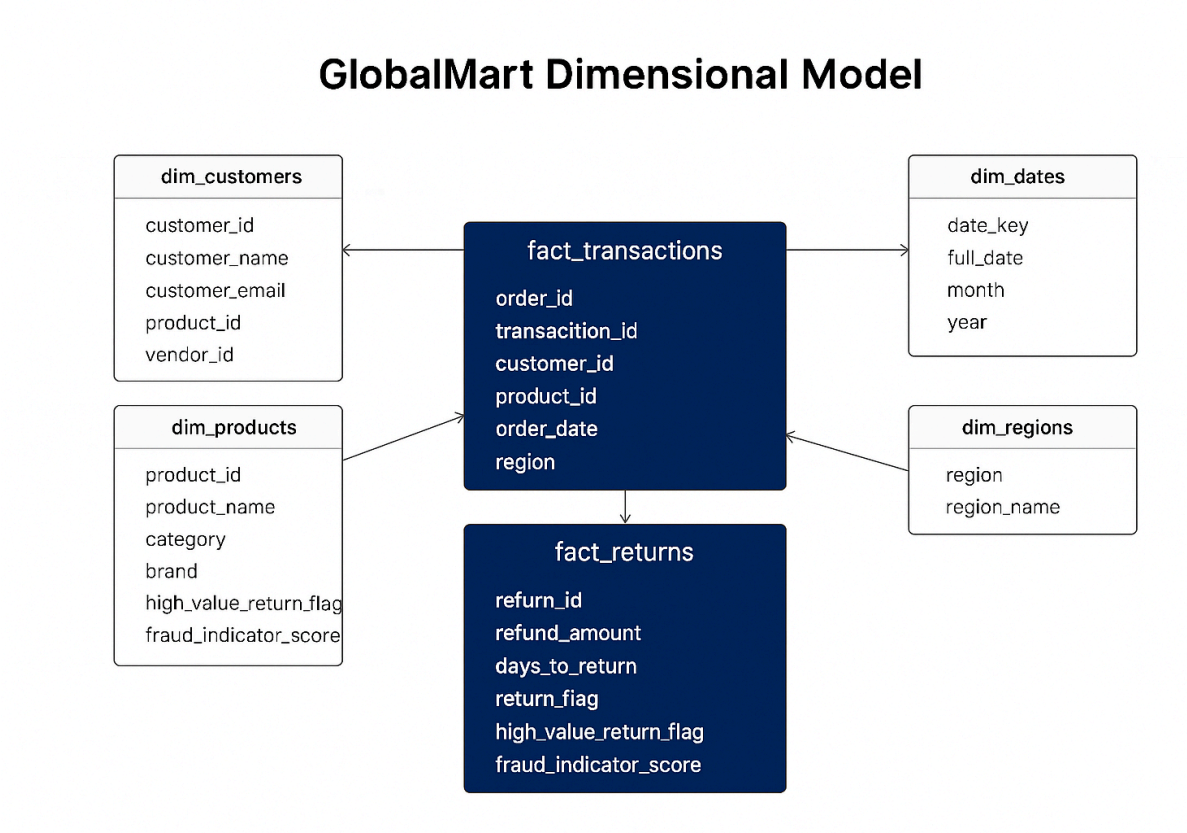
Built From (Silver Table)

Derived — from standardized region column in Silver data

Supports:

- Revenue by region
- Product performance by region
- Detecting slow-moving inventory per region

4. Relationship Diagram (Star Schema Overview)



- **Clean, simple, high-performance star schema**
- **Fully supports BI dashboards & AI semantic search**
- **All facts join cleanly via consistent keys**

5. How This Model Answers Key Business Questions

Monthly revenue by region

`fact_transactions` → `dim_dates` → `dim_regions`

Return rate by vendor

`fact_returns` + `fact_transactions` → `dim_vendors`

Customer return history across regions

`fact_returns` → `dim_customers` + `dim_regions`

Product performance by region

`fact_transactions` → `dim_products` + `dim_regions`

Fraud signals

`fact_returns` (frequency, cross-region, value)

- `fact_transactions` (order validation)
- `dim_customers` (unified identity)

6. Why This Dimensional Model Works for GlobalMart

Solves Revenue Audit Failure

- **Clear grain eliminates duplicates**
- **Fact-to-dimension mapping ensures traceability**
- **Customer identity cleaned at Silver & modeled at Gold**

Solves Return Fraud Exposure

- **fact_returns stores every return with reason + amount**
- **dim_customers enables cross-region tracking**
- **One-fact-one-event prevents inflated counts**

Solves Inventory Blindspot

- **fact_transactions provides sales velocity**
- **dim_products gives category hierarchy**
- **dim_regions breaks down regional demand**

Optimized for BI + AI

- **Lakehouse AI can answer all business questions in plain English**
- **Simple star schema makes semantic search reliable**